

Designing AI-Based Language Tools for Non-Native Speakers' Language Use and Development

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Abstract

Today's AI technology is increasingly capable of generating language resources to support human communication and language use. Yet, for non-native speakers, access to AI-generated resources does not eliminate linguistic challenges. Building upon my previous work that revealed non-native speakers' conflicting needs to manage their language production with AI tools and their communication with other people, I argue that these tools can best support non-native speakers by facilitating long-term language development in addition to assisting with specific communication tasks. In my ongoing dissertation work, I plan to use experience sampling method and participatory design to engage non-native speakers and domain experts – language educators and AI developers – in collaborative exploration towards this goal.

CCS Concepts

• Human-centered computing → Empirical studies in HCI.

Keywords

Non-Native Speakers, Language Use, Language Development, AI-Based Language Tools

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1 INTRODUCTION

AI-based language tools are increasingly capable of generating language resources to support human communication and language use (e.g., [12, 13, 18, 20, 29]). This presents an opportunity for non-native speakers who are not fully proficient in a target language to conveniently access language assistance. However, recent HCI research, including my own work, suggests that this opportunity also comes with challenges for non-native speakers in managing the language resources provided to them and can even have unintended negative impacts on their communication tasks [17, 19, 30, 31].

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Given both the opportunities and challenges associated with AI-based language tools, I argue that the promise of language support for non-native speakers lies in the integration of everyday language assistance and long-term reflective learning and language development. Following this objective, I have constructed a two-phase research plan for my dissertation that aims to 1) understand non-native speakers' needs and practices for acquiring language support in their everyday lives through an experience sampling method, and 2) identify design directions to foster learning opportunities that support non-native speakers' language development through participatory design involving them and domain experts, such as language educators and AI developers.

In this proposal, I detail the background of my research, completed studies, ongoing dissertation work, and dissertation status.

2 BACKGROUND

2.1 Language As A Resource and Non-Native Speakers' Linguistic Challenges

Language is a fundamental resource for humans to create meaning and communicate themselves with others [15, 16]. When people make an utterance, they draw upon their language resources to perform actions in the hope of producing a desired effect from their interlocutors [3, 27, 28]. Upon hearing an utterance, interlocutors also use language to offer responses and coordinate understanding with each other [9, 14, 25, 26]. Such capabilities for human interlocutors to perform social actions through language is considered a form of agency in language [2, 11].

To exercise this agency in language requires individuals to have sufficient language proficiency. For non-native speakers who lack full proficiency in a target language, this can lead to a loss of agency in their language use. For instance, in global teamwork, non-native speakers of the work language face challenges expressing their opinions and making contributions [22–24]. For immigrants who are non-native speakers of a country's dominant language, language barriers may hinder their ability to navigate daily life and access essential information in a new country [1, 10, 21]. Consequently, they often turn to tools such as translators and paraphrasers to support their language use [1, 21].

2.2 AI-Generated Language Resources as Opportunities to Support Non-Native Speakers

AI-based language tools have demonstrated the ability to generate language resources to support various domains of human communication tasks such as creative writing, journaling, and interpersonal

communication (e.g., [12, 13, 18, 20, 29]). These tools not only provide language as a form of expressions but can also proactively contribute to people's communication with their ideational capabilities. So, how can these tools assist non-native speakers who often rely on external support to navigate linguistic challenges?

A set of recent HCI studies has specifically examined how non-native speakers use these AI-based language tools (e.g., [8, 17, 19]). For instance, Ito and colleagues conducted interviews and users studies with non-native speakers with limited proficiency and found that they often used multiple tools in parallel to aid their understanding and assessment of AI-recommended texts [17]. Kim and colleagues also found that non-native speakers require support to evaluate and adopt AI-generated language suggestions and designed a paraphraser tool with multiple forms of information aids such as back translation and explanation to support this sense-making process [19]. These studies suggest that non-native speakers' limited proficiency in a target language may constrain their ability to effectively and confidently acquire language support from AI-based tools. In order to best support non-native speakers' language use with AI tools, researchers and designers need to have an in-depth understanding of non-native speakers' practices and preferences in using these tools. My research seeks to generate this empirical knowledge and derive design insights.

3 My Completed Studies on AI Tools to Support Non-Native Speakers' Language Use

During my PhD studies, I have conducted two studies involving non-native speakers to investigate their usage of AI-based language tools and their effects on language use and the communication experience. These studies revealed hidden challenges for non-native speakers in using AI tools for language support, as well as the potential for human-AI interaction design to address these challenges. These studies led to two first-authored papers accepted at ACM CSCW 2024 [30] and CHI 2025 [31], respectively. Below, I describe these studies and the core takeaways that motivate my dissertation work.

3.1 Study 1: Non-Native Speakers' AI Tool Usage in Collaborative Writing with Native Speakers [30]

We conducted an experiment to examine how non-native English speakers perform collaborative writing with native speakers. We paired dyads consisting of one non-native speaker and one native speaker to take subsequent turns and complete a joint writing task over five days. We collected participants' writing history from their document and task experience through interviews and surveys.

We found that non-native speakers frequently used AI-based writing tools to translate the content they wrote in their native language into English and to edit the English content. However, this tool usage did not fully eliminate the limitations of their English proficiency. Non-native speakers were hesitant about the language outputs generated by these tools and spent significant efforts evaluating and fine-tuning them to ensure the English content aligned with their intention. Due to the effort required to manage their language production with AI tools, non-native speakers could only

produce a limited volume of writing and concrete ideas. The non-native speakers' limited visible contributions left their co-writers frustrated. Native speakers overestimated non-native speakers' language capabilities and agency due to the relatively high quality of the content produced by non-native speakers, and were unaware of the challenges experienced by non-native speakers when producing this content.

This study highlighted a trade-off between non-native speakers' attempts to manage their language production using AI tools and their efforts to contribute meaningfully to the communication task. This trade-off, if not managed carefully, can lead to unexpected and undesired social consequences.

3.2 Study 2: Designing Human-AI Interfaces to Support Non-Native Speakers' Conversations with Native Speakers [31]

This study further investigates how non-native speakers manage two conflicting aspects of their language use: language production with AI tools and the communication process with another person. We adopted Bandura's social cognitive view of human agency that people can leverage resources to influence their behaviors and the outcomes they achieve [5, 6]. According to Bandura, human agency is dependent on the amount of resources people have: people are more likely to exercise agency with more resources, while limited resources constrain people's ability to exercise their agency [4–6].

Following this resource-agency link, we explored three human-AI interfaces to manipulate the scope of resources non-native speakers engage to produce language with AI tools. This study focused on the context of information seeking between a non-native speaking immigrant and a local person through machine translation. Specifically, the three interfaces allowed non-native speakers to do the following: 1) involve a subset of their language resources to evaluate the translation, 2) involve their full resources to evaluate and edit the translation, and 3) access English language suggestions in addition to their own resources to evaluate and edit the translation.

We conducted an experiment where non-native speakers interacted with machine translation through one of the three interfaces to converse with a native speaker. We found that non-native speakers who engaged a subset of their resources to manage language production with machine translation achieved better communication outcomes, but at the cost of reduced agency. Conversely, non-native speakers who engaged a larger scope of resources to edit machine translation outputs experienced higher agency, but at the cost of less satisfactory communication outcomes.

This finding provides additional empirical evidence that non-native speakers' use of AI-based language tools can conflict with, rather than facilitate, their communication with other people. The way non-native speakers interact with language tools to acquire language resources to support their language production impacts their communication outcomes.

Summary. Together, my two completed studies indicate that offering support for non-native speakers with AI-based language tools is not a straightforward task. While AI tools can easily provide language resources for non-native speakers, using these resources can disrupt non-native speakers' management of other aspects of their

communication. I argue that AI tools can best support non-native speakers' agency in language use by offering both task-specific language support as well as opportunities for reflective learning to enhance long-term language development. My dissertation work seeks to identify design pathways to achieve this goal.

4 ONGOING DISSERTATION WORK

Building upon my completed studies, my dissertation work explores two core research questions listed below to identify design directions for AI-based language tools to enhance non-native speakers' everyday language use and long-term language development.

RQ1. *What are the practices and desires of non-native speakers when acquiring support for everyday language use and long-term language development?*

RQ2. *How can AI-based language tools be designed to foster reflective learning opportunities for non-native speakers' long-term language development while also assisting with their everyday language use and communication tasks?*

In this section, I describe my research approach, a two-phase mixed-method research plan, and my project status.

4.1 Research Approach

I intend to ground my dissertation work in non-native speakers' everyday lived experiences of language use to complement the empirical knowledge I have acquired through controlled experiments in my previous studies. Specifically, my research plan consists of two sequential phases to: 1) acquire an understanding of non-native speakers' current practices and desires for language support (**RQ1**) through the experience sampling method, and 2) translate this understanding to AI designs (**RQ2**) through participatory design.

Phase 1. I plan to first conduct a study to sample non-native speakers' everyday language use events and their reflections on them as they occur (**RQ1**). Participants will be prompted at various points throughout a day over multiple weeks to document any communication event(s) they have recently had in their non-native language. A communication event can be a casual conversation or a thread of text message exchange, as defined by the participants themselves. For each event, participants will be instructed to share the following: 1) the context of the communication event (e.g., location, topic, and interlocutors), 2) the process and outcome of this event (e.g., how they made themselves understood, any miscommunication that occurred), 3) any preparation work they did for the event (e.g., preparing a list of possible vocabularies, translating expressions from their native language to the target language), and 4) any reflections they had about the event that (e.g., areas of improvement for their language use, useful expressions they learned from this event). I plan to analyze the data collected from this phase through thematic analysis [7] to identify emerging themes of related to communication behaviors, language support, and self-reflections on language use. These findings will generate answers to RQ1 and serve as design materials to ground the activities in Phase 2.

Phase 2. The second phase adopts a participatory design approach that involves non-native speakers and domain experts such as language educators and AI developers to collaboratively discuss

and generate design ideas (**RQ2**). I will first conduct multiple co-design workshops with non-native speakers, where they will begin by sharing their language experiences with each other, facilitated by the experience samples collected in Phase 1. Then, they will work in small groups on design activities, such as storyboarding, to illustrate how they envision AI tools supporting example communication events they have experienced and distilling language knowledge from these events to facilitate self-development. Finally, each group will share their design ideas with others and reflect on how they developed these designs and their rationales.

After the first stage of co-design workshops with non-native speakers, I will analyze my field notes for major themes regarding design considerations and collect participants' design artifacts. These will then serve as materials and prompts for the next stage of co-design workshops with domain experts. In this stage, domain experts will work in groups to generate their designs based on the same example communication events used in the first stage. They will be encouraged to leverage their pedagogical and technical expertise to generate actionable designs that foster long-term reflective learning and language development. Afterwards, I will again analyze field notes and design artifacts from these workshops, especially focusing on contrasting or complementary perspectives between non-native speakers and domain experts. Lastly, I will conduct a third stage of co-design workshops involving all three groups of participants. This stage aims to facilitate the exchange of design ideas among all parties and enable collaborative design iterations to ensure the design ideas are meaningful for non-native speakers' daily lives, pedagogically effective, and technically feasible.

4.2 Current Progress, Expected Next Steps, and Open Questions

I have developed my dissertation research plan as described in the previous section based on insights from related work and my own completed research. I have also taken two doctoral research seminars on Experience Sampling Method and Participatory Design that allowed me to gain a comprehensive understanding of these methodological approaches, acquire a collection of research techniques to implement my research, and inform my research design. As next steps, I expect to continue refining details of the study procedure, conduct pilot studies, and reach out to potential communities I plan to work with before starting data collection.

There are several open questions I continuously reflect on and hope to engage in discussions with other colleagues to gain more insights. Firstly, as I intend to ground my dissertation work in open-ended lived experience of non-native speakers, it will nevertheless introduce variability across individual participants and the data I will collect from them. I wonder: 1) What research practices or approaches can I take to account for the diversity in individual participants and leverage this diversity as resources when working with them in a longitudinal study? 2) How can I effectively navigate data that are potentially broad and complex to synthesize concrete knowledge and takeaways from my findings? Secondly, as I intend to engage participants, especially non-native speakers, in a multi-phase longitudinal study, I wonder: 3) What research designs or considerations are effective for working with participants over a longitudinal study? 4) How/if can I track the educational benefits

for participants who engage in a longitudinal study and whether this measure or reflection can become a meaningful research contribution from this study?

5 DISSERTATION STATUS AND LONG-TERM GOALS

I am currently a fourth-year PhD Candidate in the College of Information at University of Maryland, College Park. I am enrolled in a five-year full-time PhD program in Information Studies. I plan to propose my PhD dissertation by Summer 2025, and complete and defend my dissertation work by May 2026. I have never attended any doctoral consortium.

In the long-term, I plan to continue working as an HCI researcher in academic or industry positions on topics relevant to language technology, language use, and communication. My research interest is firmly grounded in my own identity and lived experience as a non-native English speaker living in an English-dominant country, and a prolonged self-exploration of my relationship with the languages I speak. I hope to continue generating empirical knowledge and design opportunities to enhance people's capacity to create meaning and express subjective experience through languages, regardless of their linguistic backgrounds. I also hope to continue collaborating with researchers and practitioners in both HCI and language-relevant fields to translate my research findings into interdisciplinary and real-world impacts.

6 CONCLUSION

My research explores the opportunities and challenges for AI-based language technology to support non-native speakers' language use. In my completed experimental studies, I found that non-native speakers have conflicting needs in managing their language production with AI tools and their communication task with other people. To mitigate this conflict, I have developed a two-phase mixed-method dissertation project to design tools that enhance non-native speakers' long-term language development beyond task-based language support. Through this project, I aim to contribute empirical knowledge and novel technology design to support non-native speakers' everyday language use and development.

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